Deep Learning–Assisted Centerline Estimation of Human Intracranial Vessels from Optical Coherence Tomography for Microsurgical Cannulation

Zhengmin Yang, MS[a,1], Daniel Chvat, BS[b,1], Alexis Lee, BS[b,1], Fiona Zhang, BS[b], Yasamin Sadat Mousavimotlagh, MD[c], Tsu-Chin Tsao, PhD[a], Aya Barzelay Wollman, MD, PhD[c,2]

[a] *Department of Mechanical and Aerospace Engineering, University of California, Los Angeles, Los Angeles, CA, USA*
[b] *Department of Electrical and Computer Engineering, University of California, Los Angeles, Los Angeles, CA, USA*
[c] *Jules Stein Eye Institute, University of California, Los Angeles, Los Angeles, CA, USA*

Abstract

Purpose: Precise vascular cannulation is critical in microsurgery but challenging due to obscured, constrained target vessels. Optical Coherence Tomography (OCT) provides high-resolution intra-operative imaging, but signal attenuation obscures the posterior vessel wall, preventing automated 3D targeting.

Design: Experimental study

Subjects, Participants, and/or Controls: Human cadaveric specimens were used to expose the internal carotid artery (ICA), anterior cerebral artery (ACA), and ophthalmic artery (OA) for OCT imaging as part of anatomical investigations supporting whole-eye transplantation strategies aimed at achieving zero warm ischemia through retrograde extracorporeal perfusion[1].

Methods: We propose a deep learning and geometric reconstruction pipeline to estimate the anatomical centerline and caliber of human intracranial and ophthalmic vessels from incomplete OCT volumes. A ConvGRU bottleneck maintains temporal consistency across B-scans, while curvature-based projection localizes the vessel center. The OCT volumes used for training were extracted from two donors.

Main Outcome Measures: Estimated vessel diameters compared with manual annotations and literature reference diameters.

Results: Tested on human cadaveric specimens, the framework successfully extracted vessel

Email address: ABarzelayWollman@mednet.ucla.edu (Aya Barzelay Wollman, MD, PhD)

[1]These authors contributed equally to this work.

[2]Corresponding author. Email: ABarzelayWollman@mednet.ucla.edu; Tel.: +1-310-206-7474.

parameters. RANSAC outlier rejection retained 89.5% of internal carotid artery (ICA), 86.1% of anterior cerebral artery (ACA), and 45.8% of ophthalmic artery (OA) slices. Estimated diameters revealed post-mortem deformations: ICA contracted by 36.8%, while OA expanded by 21.1% due to dural tethering.

Conclusions: This framework enables robust 3D vascular localization from partial OCT data, providing quantitative geometric feedback to augment surgical judgment.

Keywords: Optical coherence tomography, deep learning, centerline estimation, microsurgical cannulation, whole-eye transplantation

---

Word Count: 2961

Figures: 4

Tables: 1

Financial Support: This work was supported by the Advanced Research Projects Agency for Health (ARPA-H) Transplantation of Human Eye Allografts (THEA) program. The sponsor or funding organization had no role in the design or conduct of this research.

Conflict of Interest: No conflicting relationship exists for any author

Abbreviations and Acronyms: CA = anterior cerebral artery; AI = artificial intelligence; ARPA-H = Advanced Research Projects Agency for Health; CNN = convolutional neural network; ConvGRU = convolutional gated recurrent unit; ICA = internal carotid artery; MCA = middle cerebral artery; OA = ophthalmic artery; OCT = optical coherence tomography; RANSAC = random sample consensus; R-UNet = recurrent U-Net; T-BPTT = truncated backpropagation through time; THEA = Transplantation of Human Eye Allografts.

Introduction

The Advanced Research Projects Agency for Health (ARPA-H) Transplantation of Human Eye Allografts (THEA) program aims to advance whole-eye transplantation through rapid donor eye procurement, extracorporeal perfusion, and restoration of ocular viability . In these settings, surgeons must establish vascular access to small ophthalmic and intracranial vessels located near the optic canal and skull base, where direct visualization is limited and optical coherence tomography (OCT) is increasingly integrated into ophthalmic microsurgery because of its ability to provide micron-scale intraoperative visualization of delicate ocular and orbital anatomy. Emerging ophthalmic applications involving vascular access and vessel localization include intra-arterial ophthalmic artery catheterization for retinoblastoma chemotherapy, orbital vascular embolization procedures, skull-base and orbital tumor surgery, and image-guided robotic microsurgical interventions . Under OCT scans, vessel geometry may be only partially visible intraoperatively. Accurate identification of the vascular centerline is therefore critical for safe needle guidance and instrument trajectory planning, as trajectory misalignment may lead to posterior wall puncture, intramural dissection, thrombosis, or failed vascular access.

Although routinely performed, vascular cannulation remains geometrically challenging because target vessels are small, deformable, and frequently partially obscured within constrained surgical corridors. These complications persist despite image guidance, with posterior wall puncture rates of 17–40% reported during ultrasound-guided vascular cannulation due to incomplete visualization of the needle tip and vessel lumen[2–5].

Automation and robot-assisted surgery offer significant benefits for these constrained procedures, but their success hinges on robust 3D vessel localization. Current automated centerline detection methods primarily focus on 2D projections using Hessian-based filters or foundational deep learning architectures such as U-Net and R2U-Net[6,7]. While effective in 2D, these approaches lack the cross-sectional depth and volumetric continuity required for high-precision 3D surgical guidance. Conversely, 3D macro-imaging modalities such as ultrasound and computed tomography (CT) assume that the entire vessel cross-section is identifiable. Existing robotic vascular access systems rely on this assumption of complete contour visibility[8,9]. However, macro-imaging lacks the micron-scale axial resolution necessary for microvascular interventions in ophthalmic and intracranial surgery, where tortuous and tethered vessels demand higher precision.

Augmented intraoperative imaging, specifically OCT, partially addresses this limitation by providing high-resolution, depth-resolved cross-sectional visualization during surgery[10–12]. Prior studies have utilized OCT for quantitative vascular characterization, including vessel wall measurements and vessel classification[13,14]. However, interventional OCT introduces a fundamental limitation for vascular access: signal attenuation and limited penetration depth, particularly in highly scattering tissue environments, commonly obscure the posterior vessel wall and allow only the superficial vessel boundary to be visualized[15]. The posterior wall is therefore frequently hidden by optical shadowing, rendering traditional full-contour segmentation methods ineffective.

Currently, there is no established method for determining the centerline of a blood vessel from incomplete cross-sectional OCT images or utilizing known anatomical diameters to reconstruct missing vascular geometry for surgical guidance. Maintaining axial consistency across sequential B-scans is notoriously difficult in volumetric OCT, although recurrent neural network architectures have demonstrated promise in preserving spatial continuity for retinal layer segmentation[7].

In this work, we address this limitation by estimating vascular centerline and caliber from OCT volumes even when only the superior vessel boundary is reliably visible. The proposed framework combines recurrent deep-learning segmentation, utilizing a convolutional gated recurrent unit (ConvGRU) bottleneck to maintain axial continuity despite severe shadowing, with curvature-based geometric reconstruction to estimate the anatomical vessel center from incomplete vascular boundaries. This approach generates stable 3D centerline representations that may support future image-guided and robot-assisted microsurgical interventions.

Using OCT volumes acquired from human cadaveric ophthalmic and intracranial vessels, we demonstrate the feasibility of reconstructing stable vessel geometry from partial intraoperative imaging data. Compared to human determined blood vessel diameters from the same scans, our automated approach predicted diameters that were closer to the expected values compared to a manual approach. Rather than replacing surgical decision-making, this approach aims to augment surgical judgment by providing quantitative geometric information regarding vessel location and stability, representing a critical step toward image-guided cannulation in advanced microsurgery.

Methods

The proposed pipeline for the automated localization and segmentation of ocular blood vessel top boundaries in OCT B-scans consists of three primary stages: (1) a binary classification gatekeeper to manage data quality and mitigate the effects of class imbalance, (2) a recurrent convolutional neural network (R-UNet) for temporally consistent segmentation of the blood vessel boundaries and peaks, and (3) a geometric post-processing stage for anatomical center estimation from the segmented blood vessel boundary.

Cadaveric Specimen Preparation

Two fresh human cadavers were used to expose the internal carotid artery (ICA), anterior cerebral artery (ACA), middle cerebral artery (MCA), and ophthalmic artery (OA) for OCT imaging as part of anatomical investigations supporting whole-eye transplantation strategies aimed at achieving zero warm ischemia through retrograde extracorporeal perfusion[1]. All dissections were performed under institutional anatomic research protocols and approved by the University of California, Los Angeles Anatomy Lab. A modified fronto-temporo-orbito-zygomatic craniotomy was performed to create a wide surgical corridor to the optic canal, cavernous sinus, and proximal circle of Willis[16,17]. Using standard microsurgical techniques, the optic chiasm and the A1 segment of the ACA together with the proximal M1–M2 segments of the middle cerebral artery were exposed. The distal dural ring was incised to visualize the cavernous segment of the ICA and the origin of the ophthalmic artery near the optic canal. Adjacent cranial nerves and the superior ophthalmic vein were identified to maintain anatomical orientation during the dissection. This approach provided stable exposure of the ICA, ACA, and OA while preserving the surrounding skull-base anatomical relationships required for OCT imaging.

Data Acquisition

8 OCT C-scans, of various ocular vessels (two ophthalmic artery scans, two internal carotid artery scans, one middle cerebral artery scan, and three anterior cerebral artery scans), each with 128 B-scans, were acquired using the Zeiss Opti Lumera 7000. The dimensions of the scan were 10mm by 10mm by 2.8mm. The C-scan was rotated such that the central B-scan is aligned with the blood vessel of interest and that the vessel of interest is centered in the B-Scan depth.

The authors labelled each B-scan image. The blood vessel outline and top center of the blood vessel were labelled. The top center labelling is used to obtain the point of local minima for the geometric fitting that will be addressed in the Geometric Center Estimation section. The labelling was validated by other labellers.

Due to the limited data, blood vessels from both specimens were present in both the training and test data. In addition, to create more data, we performed randomized image augmentation (Gaussian noise, colour jitter, elastic transform, random brightness contrast, affine transform) for each C-scan. A total of 5 augmented C-scans were created from each original C-scan, producing 48 total C-scans used to train the model.

Core Assumptions

To bridge the gap between the collapsed experimental data and the pressurized clinical target and to demonstrate method feasibility, the following assumptions were established:

- **Scan Orientation:** OCT B-scans were acquired approximately perpendicular to the longitudinal vessel axis, allowing each slice to be treated as a transverse vascular cross-section. The vessel of interest was positioned near the center of the scan volume during acquisition. Minor deviations in alignment were addressed during post-processing using RANSAC-based outlier rejection.

- **Vessel Geometry:** Vascular cross-sections were modeled using locally circular geometry to estimate vessel curvature and center position. This approximation was used to reconstruct the anatomical centerline from partial vessel boundary information.

- **Superior Boundary Stability:**The superficial (superior) vessel boundary was used as the primary geometric feature for center estimation because it remained consistently visible across OCT scans, whereas the posterior vessel wall was frequently obscured by signal attenuation and optical shadowing.
- **Volumetric Continuity:** Sequential B-scans were assumed to contain spatially continuous vascular morphology, enabling recurrent neural network modeling of vessel geometry across the OCT volume.
- **OCT Image Inversion:** OCT Image Orientation: OCT image inversion correction was not included in the present framework because established image-processing and deep learning methods for inversion correction have been previously described[18,19].

Multi-Stage Network Pipeline

A multi-stage deep learning pipeline (Figure 1) was adopted to estimate the blood vessel centerline from OCT images. This helped to mitigate the large class imbalance between the background class and the classes we are interested in, namely the blood vessel wall and the blood vessel peak. Since there could be many "peaks" in the image due to surrounding tissue artifacts, it is necessary to identify the peak that is associated with the blood vessel. The blood vessel boundary and peak information allows for the estimation of the blood vessel centerline while only seeing the top half of the blood vessel in the OCT scan. Deep learning also learns the noise in the OCT images, enhancing the detection quality versus traditional image processing.

Pre-filtering Stage (EmptyMaskCNN)

An *EmptyMaskCNN* binary classifier was implemented to exclude B-scans lacking identifiable vascular structures from downstream processing. The architecture comprises three downsampling blocks (16, 32, and 64 filters) utilizing 3×3 convolutions and 2×2 max-pooling. This gatekeeper ensures that the segmentation network only processes signal-rich slices, effectively managing class imbalance.

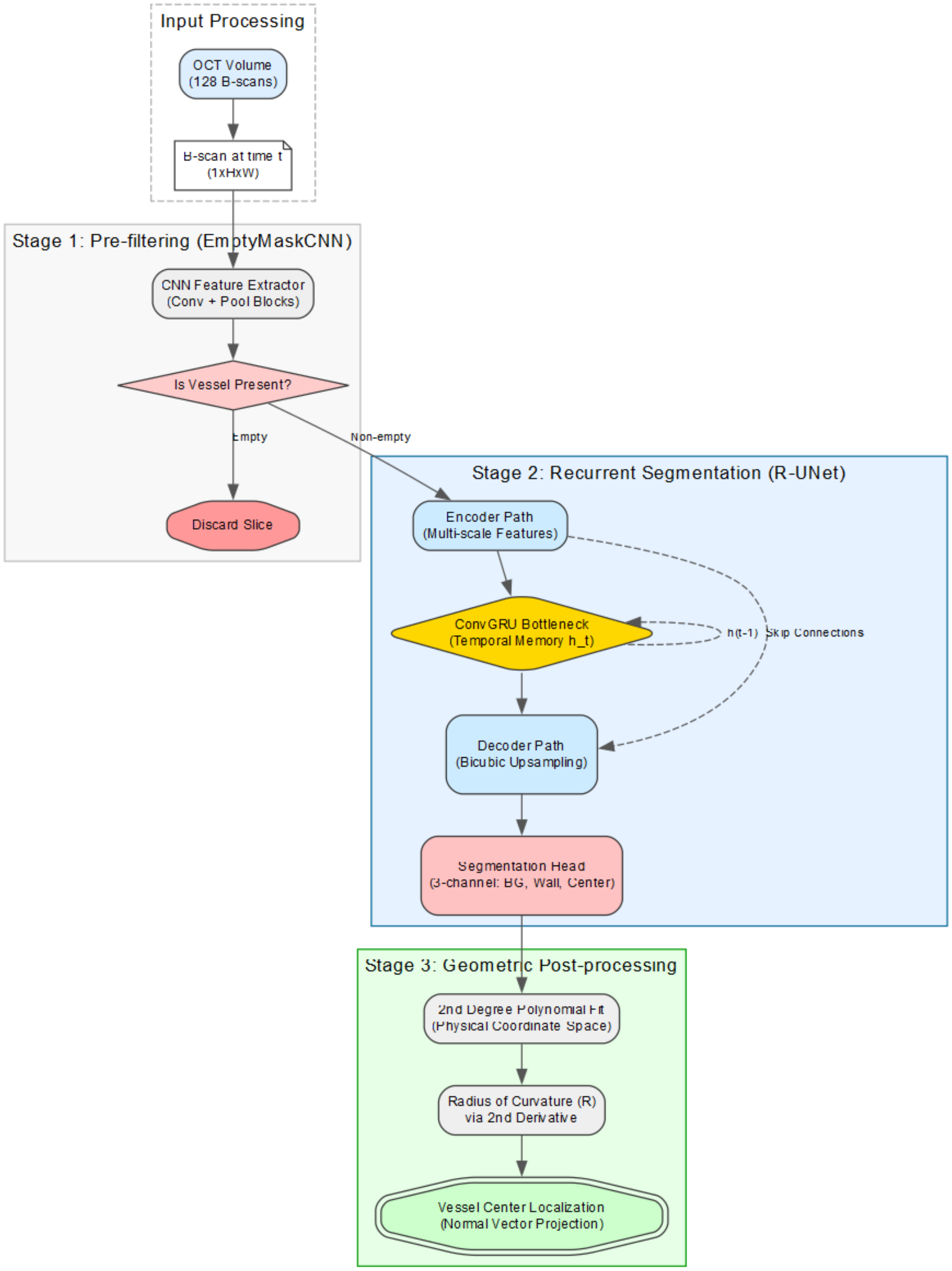


Figure 1: Architecture of the entire pipeline. The pipeline is composed of three stages: the Pre-filtering Stage, Segmentation Network, and Geometric Center Estimation.

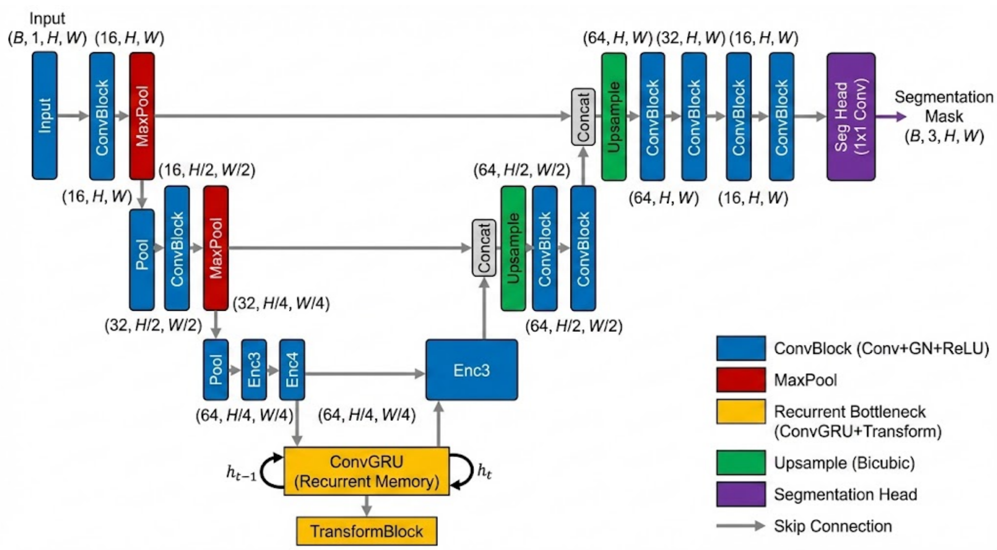


Figure 2: Architecture of the Recurrent U-Net (R-UNet). The ConvGRU at the bottleneck maintains a spatial hidden state across sequential B-scans to ensure temporal consistency of the segmented boundaries.

Recurrent Segmentation Network (R-UNet)

The core segmentation is performed by a customized U-Net variant (Figure 2) utilizing a Convolutional Gated Recurrent Unit (ConvGRU) at the bottleneck ($H/4, W/4$ resolution).

- **Temporal Modeling:** The ConvGRU propagates a hidden state h_t between successive B-scans to track vessel morphology across the volume, ensuring axial continuity.
- **Loss:** To address the severe class imbalance between the background class and the sparse target features (vessel wall and peak), the network was optimized using a weighted sum of Focal Tversky Loss[20] and Dice Loss[21].
- **Training Protocol:** The model was trained for 30 epochs with a cosine annealing learning rate scheduler (1e-3 initial learning rate) and the Adam optimizer on an NVIDIA GeForce RTX 4080 Ti . A Truncated Backpropagation Through Time (T-BPTT) strategy was implemented with a window of 20 slices, representing the maximum sequence length permissible by hardware VRAM constraints.

Geometric Center Estimation

Following R-UNet segmentation, raw masks are mapped to a physical coordinate space (10.0 × 2.8 mm). A second-degree polynomial was fitted to the identified top boundary. This was chosen because, while blood vessel cross sections are elliptical, the B-scans produce partially occluded blood vessels that only show at most the upper half of an ellipse. The standard equation for this, assuming the ellipse is centered at (x_c, y_c) with semi-major and semi-minor axes a and b, is

$$y(x) = y_c + b\sqrt{1 - \frac{(x - x_c)^2}{a^2}} \tag{1}$$

Near the apex, $\frac{x - x_c}{a}$ is small. Applying the binomial approximation $\sqrt{1-u} \approx 1 - \frac{u}{2}$ yields:

$$y(x) \approx (y_c + b) - \frac{b}{2a^2}(x - x_c)^2 \tag{2}$$

This simplifies to $y(x) = A(x - x_c)^2 + K$, which is a second-order polynomial.

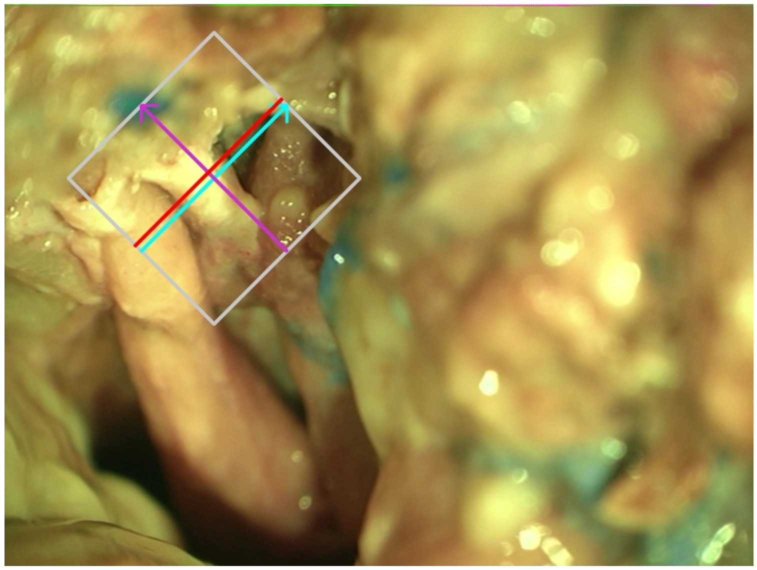
By calculating the radius of curvature R using the second derivative:

$$R = \frac{(1 + (f'(x))^2)^{1.5}}{|f''(x)|} \tag{3}$$

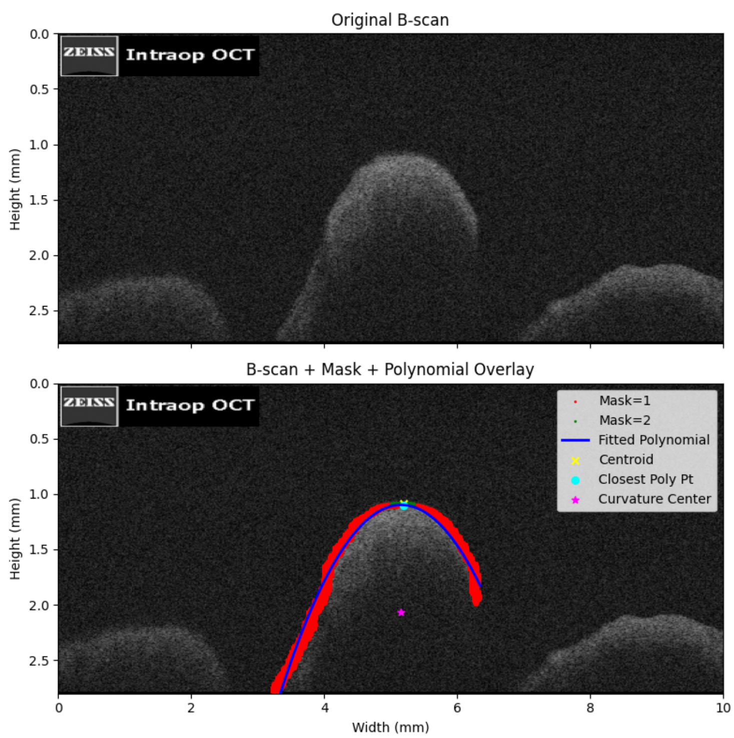
The point at which R was calculated was identified by the top center label. The anatomical center is localized by projecting along the normal vector from the top boundary into the vessel lumen by distance R.

RANSAC Outlier Removal

RANSAC was used to remove any erroneous detections to improve the robustness of the detection. The rejection threshold (i.e. max deviation between retained samples) is set at 0.1 mm to prioritize estimation precision over retention of imprecise data.



(a)



(b)

Figure 3: (a) Microscope view of the OA section scanned by the OCT; the example slice is shown in red. (b) Geometric reconstruction process: A second-degree polynomial (blue) is fitted to the segmented boundary (red). The radius of curvature is used to define an osculating circle, the center of which (magenta) represents the estimated anatomical vessel center.

Results

Vessel Centerline Estimation and Outlier Rejection

The proposed curvature-based reconstruction framework was applied to OCT volumes of the Internal Carotid Artery (ICA), Anterior Cerebral Artery (ACA), and Ophthalmic Artery (OA). RANSAC-based filtering was used to remove segmentation inconsistencies and geometric outliers prior to centerline estimation. Stable vessel centerline reconstruction was achieved across sequential OCT B-scans in all evaluated vessel groups following outlier rejection. The framework demonstrated lower geometric variability in the larger intracranial vessels. Following RANSAC filtering, 31 of 36 ACA slices (86.1%) and 17 of 19 ICA slices (89.5%) were retained. In comparison, the OA demonstrated greater variability, with 11 of 24 slices (45.8%) retained following outlier rejection.

Quantitative Diameter Analysis

We also had a human volunteer hand label the radii from the estimated center to the lumen wall. Scans that did not show at least half the vascular cross section were ignored similar to the pre-filtering stage of our model. This was done to allow for a direct comparison. Table 1 summarizes our machine learning assisted estimated diameters compared to human estimated diameters and ground truth values derived from literature (OA:[22], ACA:[23], ICA:[24]).

- **Internal Carotid Artery (ICA):** The mean estimated diameter was 2.94 ± 0.17 mm. This represents a deviation of -36.8% from the expected ground truth of 4.66 mm.
- **Anterior Cerebral Artery (ACA):** The mean estimated diameter was 1.64 ± 0.45 mm, showing a deviation of -15.2% from the ground truth of 1.93 mm.
- **Ophthalmic Artery (OA):** The mean estimated OA diameter was 2.12 ± 0.37 mm, exhibiting a positive deviation of $+21.1\%$ compared to the ground truth of 1.75 mm.

Table 1: Comparison of Estimated Vessel Diameters vs. Literature Ground Truth

Vessel	**ML Est. Diameter (mm)**	**Human Est. Diameter (mm)**	**Ground Truth (mm)**	**% Deviation Human**	**% Deviation ML**
ICA	2.94 ± 0.17	1.674 ± 0.270	4.66 ± 0.78	64%	-36.8%
ACA	1.64 ± 0.45	1.029 ± 0.356	1.93 ± 0.93	33%	-15.2%
OA	2.12 ± 0.37	1.180 ± 0.340	1.75 ± 0.25	47%	$+21.1\%$

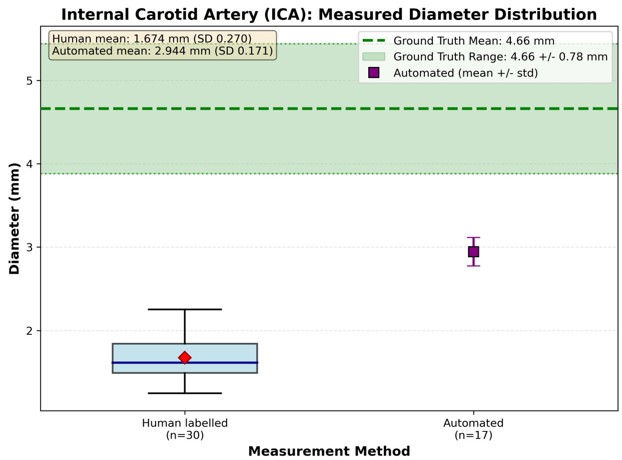
Figure 4 illustrates the distribution of estimated vessel diameters across all evaluated vascular segments.

Discussion

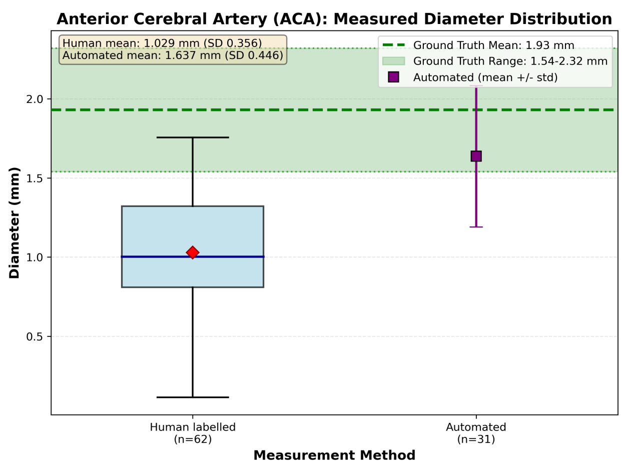
Whole eye transplantation has recently emerged as a potential strategy for restoring vision in patients with severe ocular trauma or irreversible optic nerve injury[25–27]. Recent work by our group has demonstrated the feasibility of maintaining zero warm ischemia during donor eye procurement through retrograde extracorporeal perfusion of the ophthalmic circulation[1]. A critical technical challenge in this procedure is the rapid establishment of vascular access to preserve and perfuse the donor eye following procurement. Cannulation of the ophthalmic artery and associated intracranial vessels is therefore essential for maintaining tissue viability prior to transplantation. However, these vessels are small, tortuous, and located within constrained anatomical corridors near the skull base and optic canal, where direct visualization of the vessel lumen is limited.

The proposed reconstruction framework demonstrated differing levels of geometric stability across the evaluated vascular segments. The ICA and ACA exhibited relatively consistent centerline reconstruction and high slice retention following RANSAC filtering, whereas the OA demonstrated substantially greater geometric variability. These findings are consistent with the anatomical complexity emphasized in the Introduction, where ophthalmic and orbital microsurgical procedures frequently involve small, tortuous vessels located within constrained surgical corridors near the optic canal and skull base[28–30]. The increased variability observed in the OA likely reflects the vessel's curved intraorbital course, variable caliber, and local anatomical tethering. Together, these findings suggest that vascular tortuosity and regional anatomical constraints may significantly influence OCT-based centerline reconstruction performance in small ophthalmic vessels.

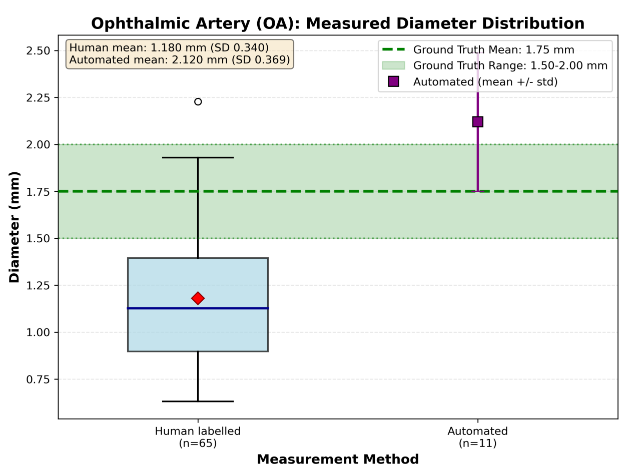
Manual annotation validation further supported the anatomical plausibility of the proposed reconstruction framework. Agreement between expert-derived center estimates and algorithm-generated centerlines suggests that meaningful vascular geometry can be recovered despite incomplete visualization of the posterior vessel wall. The higher agreement observed in the ICA and



(a)



(b)



(c)

Figure 4: Box and whisker plots comparing the estimated vessel diameters against literature ground truth ranges. Note the tighter variance in the ICA and ACA compared to the OA.

ACA compared with the OA is also consistent with the reduced geometric variability and higher slice retention demonstrated in the Results section. These findings support the feasibility of combining recurrent OCT segmentation with curvature-based geometric reconstruction for quantitative vascular localization in microsurgical imaging environments.

The observed reductions in ICA and ACA diameter measurements are consistent with known post-mortem biomechanical changes in intracranial vasculature. The ICA demonstrated the greatest reduction relative to literature reference values, which may reflect contraction of muscular arterial tissue following circulatory collapse and tissue fixation. Histological studies demonstrate that the cavernous and supraclinoid ICA segments exhibit characteristics of muscular arteries and lack a distinct external elastic lamina[31]. Experimental ex vivo vascular studies have demonstrated substantial lumen reduction following post-mortem smooth muscle contraction[32]. In contrast, the ACA demonstrated more moderate geometric reduction, potentially due to structural support provided by surrounding arachnoid architecture and fixation-related dimensional changes[33,34].

Unlike the intracranial vessels, the OA demonstrated larger measured diameters relative to angiographic reference values. Several anatomical and imaging-related factors may contribute to this finding. The ophthalmic artery is closely associated with the dural sheath of the optic nerve as it traverses the optic canal[30], which may provide structural reinforcement and reduce post-mortem collapse. In addition, OCT imaging captures vessel wall geometry rather than lumen-only measurements obtained by angiographic modalities such as CTA or DSA. These differences may contribute to the increased diameter measurements observed in the OA[35,36].

The primary goal of this work is not to automate vascular cannulation, but rather to provide quantitative geometric information that may augment surgical judgment during microsurgical procedures. In ophthalmic and orbital microsurgery, surgeons frequently infer vessel trajectory from limited visual cues within constrained anatomical environments. Although OCT provides high-resolution cross-sectional imaging, individual OCT slices do not directly reveal the three-dimensional orientation of the vessel[10,15]. By reconstructing vessel centerlines from sequential OCT B-scans, the proposed framework converts partial cross-sectional observations into coherent three-dimensional vascular geometry that may assist surgeons in identifying suitable cannulation segments and estimating safe insertion trajectories.

Although motivated by ophthalmic transplantation and microsurgical vascular access, similar imaging challenges arise across a broad range of microsurgical and reconstructive procedures involving small vessels located within complex anatomical environments. The ability to reconstruct vessel centerlines from incomplete OCT observations may therefore have broader implications for intraoperative imaging, augmented surgical visualization, and future robotic guidance systems. OCT has already demonstrated substantial promise for intraoperative imaging because of its micron-scale spatial resolution[10,15]. Integrating OCT imaging with geometric reconstruction methods may allow surgeons to visualize vessel curvature, branching geometry, and regional diameter variation during microsurgical procedures.

Importantly, this study demonstrates that meaningful vascular geometry can be reconstructed directly from OCT volumes acquired from human intracranial and ophthalmic vessels rather than relying exclusively on synthetic phantoms or animal models. This strengthens the translational relevance of the proposed approach and supports the feasibility of OCT-based geometric reconstruction as a potential component of future image-guided microsurgical systems.

Limitations

Several limitations should be considered when interpreting these findings. First, the study was performed using ex vivo human cadaveric specimens, and vascular geometry may therefore differ from perfused in vivo anatomy because of post-mortem tissue changes and the absence of physiologic blood pressure. However, the use of human intracranial and ophthalmic vessels allowed evaluation within anatomically relevant microsurgical environments that are difficult to replicate using synthetic phantoms or animal models. Future studies may include perfused human cadaveric model.

Neural network training was limited by the number of human specimens, which restricted both the size and diversity of the dataset. Additional data would likely improve the robustness and generalizability of the neural network, particularly across variable vascular anatomy and imaging conditions.

In addition, quantitative validation was performed on a limited subset of manually annotated OCT B-scans, and larger-scale validation studies may further characterize reconstruction performance across broader vascular anatomies and imaging conditions. Despite these limitations, the proposed framework demonstrated stable centerline reconstruction across multiple anatomically relevant vascular segments and supports the feasibility of OCT-based geometric reconstruction for future image-guided microsurgical applications.

In this work, we presented a multi-stage deep learning and geometric reconstruction framework to estimate the anatomical centerline and caliber of intracranial and ophthalmic blood vessels from incomplete OCT volumes. By utilizing a ConvGRU bottleneck to enforce temporal consistency across sequential B-scans and applying curvature-based geometric projection, the system successfully extracted stable 3D centerline representations even when only the superior vessel boundary was visible. Experimental application on ex-vivo human cadaveric specimens demonstrated the robustness of this approach across varying vessel morphologies. Furthermore, the method successfully estimated vessel centers, revealing physiological deformations consistent with post-mortem mechanics. The data suggests that future cannulation planning must account for significant caliber loss in the carotid system, while anticipating a structurally preserved target in the OA. Ultimately, this framework converts partial cross-sectional OCT observations into actionable quantitative data, augmenting surgical judgment and establishing a critical foundation for image-guided and robot-assisted cannulation in advanced microsurgery. Future work could be done on looking at higher order polynomial degree fits for determining the centerlines of blood vessels which are side-by-side, as well as further enhancements on the machine learning model.

Acknowledgments

The authors wish to thank individuals who donate their bodies and tissues for the advancement of education and research.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors utilized Google Gemini (Gemini Advanced, Google LLC) to assist with editing wording and improving sentence flow. The authors thoroughly reviewed, edited, and verified all AI-assisted text to ensure accuracy and maintain the original scientific meaning. The authors take full responsibility for the final content of this manuscript.

References

[1] A. Wang, et al., Zero warm ischemia time in whole-eye transplantation through retrograde extracorporeal perfusion, Ophthalmology ScienceAccepted for publication (2026).

[2] C. Moon, D. Blehar, M. Shear, P. Uyehara, R. Gaspari, J. Arnold, J. Cukor, Incidence of posterior vessel wall puncture during ultrasound-guided vessel cannulation in a simulated model, Academic Emergency Medicine 17 (10) (2010) 1138–1141.

[3] N. Kumar, A. Bindra, G. Singh, K. Goyal, A. Chaturvedi, N. Yadav, et al., Incidence of posterior vessel wall puncture during ultrasound-guided vascular access: short-axis versus long-axis approach, Journal of Anaesthesiology Clinical Pharmacology 37 (3) (2021) 342–346.

[4] C. Dietrich, R. Horn, S. Morf, L. Chiorean, Y. Dong, X. Cui, Ultrasound-guided central vascular interventions, Journal of Thoracic Disease 8 (12) (2016) E851–E868.

[5] S. Srinivasan, D. Govil, S. Gupta, S. Patel, A. Jagadeesh, D. Tomar, Incidence of posterior wall penetration during internal jugular vein cannulation: A comparison of two techniques using ultrasound, Indian Journal of Anaesthesia 61 (9) (2017) 772–778.

[6] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Springer, 2015.

[7] M. Z. Alom, et al., Recurrent residual convolutional neural network based on u-net (r2u-net) for medical image segmentation, arXiv preprint arXiv:1802.06955 (2018).

[8] R. D. Brewer, J. K. Salisbury, Visual vein-finding for robotic iv insertion, in: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), IEEE, Anchorage, Alaska, USA, 2010, pp. 4597–4602.

[9] Y.-C. Chang, B. Berry-Pusey, R. Yasin, N. Vu, B. Maraglia, A. X. Chatziioannou, T.-C. Tsao, An automated mouse tail vascular access system by vision and pressure feedback, IEEE/ASME Transactions on Mechatronics 20 (4) (2015) 1616–1623. doi:10.1109/TMECH.2014.2360886.

[10] O. Carrasco-Zevallos, C. Viehland, B. Keller, M. Draelos, A. Kuo, C. Toth, J. Izatt, Review of intraoperative optical coherence tomography: technology and applications, Biomedical Optics Express 8 (3) (2017) 1607–1637.

[11] E. Chatzimichail, G. Chondrozoumakis, F. Doroodgar, L. Ham, I. Dapena, G. Melles, Emerging applications of intraoperative optical coherence tomography in corneal surgery: a narrative review, Journal of Clinical Medicine 13 (18) (2024) 5426.

[12] D. Cheng, Y. Li, R. Wang, Intraoperative optical coherence tomography angiography: principles and applications, Frontiers in Physics 11 (2023) 1145882.

[13] T. H. Rim, et al., Quantitative analysis of retinal vessel morphology using spectral-domain optical coherence tomography, OphthalmologyNIHMSID: nihms727709 (2016).

[14] T. e. a. ADEJUMO, Automatic segmentation of retinal blood vessels in oct images based on a deep learning approach, Biomedical Optics Express 14 (12) (2023) 6350–6365. doi:10.1364/BOE.501530.

[15] Y. Tao, O. Carrasco-Zevallos, J. Ehlers, C. Toth, J. Izatt, Intraoperative optical coherence tomography in ophthalmology, Progress in Retinal and Eye Research 94 (2025) 101164.

[16] J. M. Zabramski, T. Kiriş, S. K. Sankhla, R. F. Spetzler, Orbitozygomatic craniotomy, Journal of Neurosurgery 89 (2) (1998) 336–346.

[17] G. M. Lemole, J. S. Henn, J. M. Zabramski, R. F. Spetzler, Modifications to the orbitozygomatic approach, Journal of Neurosurgery 99 (5) (2003) 924–930.

[18] V. Bellemo, R. Haindl, M. Pramanik, L. Liu, L. Schmetterer, X. Liu, Complex conjugate removal in optical coherence tomography using phase aware generative adversarial network, Journal of Biomedical Optics 30 (2) (2025) 026001.

[19] X. Attendu, D. J. Faber, G. Lamouche, T. G. van Leeuwen, C. Boudoux, M. Rivard, Calibration procedure for enhanced mirror artifact removal in full-range optical coherence tomography using passive quadrature demultiplexing, Journal of Biomedical Optics 27 (11) (2022) 116006.

[20] N. Abraham, N. M. Khan, A novel focal tversky loss function with improved attention u-net for lesion segmentation, in: 2019 IEEE 16th international symposium on biomedical imaging (ISBI 2019), IEEE, 2019, pp. 683–687.

[21] F. Milletari, N. Navab, S.-A. Ahmadi, V-net: Fully convolutional neural networks for volumetric medical image segmentation, in: 2016 fourth international conference on 3D vision (3DV), IEEE, 2016, pp. 565–571.

[22] V. Iulia, B. Petru, I. Viorel, Considerations on the origin of the ophthalmic artery from the internal carotid artery, ARS Medica Tomitana 24 (2018) 179–183. doi:10.2478/arsm-2018-0037.

[23] A. Karatas, G. Coban, C. Cinar, I. Oran, A. Uz, Assessment of the circle of willis with cranial tomography angiography, Medical Science Monitor: International Medical Journal of Experimental and Clinical Research 21 (2015) 2647, source for ACA A1 segment morphometry (1.93-1.96 mm).

[24] J. Krejza, M. Arkuszewski, S. E. Kasner, J. Weigele, A. Ustymowicz, R. W. Hurst, B. L. Cucchiara, S. R. Messe, Carotid artery diameter in men and women and the relation to body and neck size, Stroke 37 (4) (2006) 1103–1105, source for ICA ground truth (4.66 mm female mean).

[25] M.-C. Bekono-Nessah, A. Scarabosio, M. Laspro, K. K. Gokoffski, A. M. Al-Mousawi, Whole-eye transplantation: How far are we from a breakthrough?, Ophthalmic Plastic and Reconstructive Surgery 40 (6) (2024) 601–608. doi:10.1097/IOP.0000000000002738.

[26] M. Laspro, B. F. Chaya, H. T. Brydges, N. Dave, E. Thys, O. C. Onuh, D. Tran, L. L. Kimberly, D. J. Ceradini, E. D. Rodriguez, Technical feasibility of whole-eye vascular composite allotransplantation: A systematic review, Plastic and Reconstructive Surgery Global Open 11 (4) (2023) e4946. doi:10.1097/GOX.0000000000004946.

[27] K. K. Gokoffski, D. J. Ceradini, R. M. Siatkowski, Clinical and scientific considerations for whole eye transplantation: An ophthalmologist's perspective, Translational Vision Science & Technology 14 (2) (2025) 13. doi:10.1167/tvst.14.2.13.

[28] Y. P. Gobin, I. J. Dunkel, B. P. Marr, et al., Intra-arterial chemotherapy for the management of retinoblastoma: four-year experience, Arch Ophthalmol 129 (6) (2011) 732–737.

[29] T. A. Abruzzo, H. A. Riina, M. W. Kieran, et al., Superselective ophthalmic artery infusion of chemotherapy for treatment of retinoblastoma, AJNR Am J Neuroradiol 33 (8) (2012) 1608–1614.

[30] S. S. Hayreh, Orbital vascular anatomy, Eye 20 (10) (2006) 1130–1144, definitive source on OA dural sheath relationships and muscular wall histology.

[31] T. Masuoka, N. Hayashi, E. Hori, N. Kuwayama, O. Ohtani, S. Endo, Distribution of internal elastic lamina and external elastic lamina in the internal carotid artery: Possible relationship with atherosclerosis, Neurologia medico-chirurgica 50 (3) (2010) 179–182, definitive source

confirming the intracranial ICA is a MUSCULAR artery that lacks the External Elastic Lamina.

[32] M. Buscema, S. E. Hieber, G. Schulz, et al., Ex vivo evaluation of an atherosclerotic human coronary artery via histology and high-resolution hard x-ray tomography, Scientific Reports 9 (1) (2019) 1–10, discusses 50-80% lumen reduction in ex vivo states.

[33] H. Kretschmann, W. Weinrich, The shrinkage of the human brain stem during formalin fixation and embedding in paraffin, Brain Structure and Function (Anatomy and Embryology) 174 (1997) 419–429, the definitive source citing 11-17% shrinkage for intracranial structures due to formalin alone.

[34] K. Zhao, Y. Li, et al., Micro-anatomic study of the arachnoid trabeculae in the cranial subarachnoid space, Surgical and Radiologic Anatomy 37 (2015) 79–86, supports the claim that arachnoid trabeculae tether intracranial vessels, preventing total collapse.

[35] J. Lang, Clinical Anatomy of the Head: Neurocranium and orbit, Springer-Verlag, Berlin, Heidelberg, 1983, classic anatomical text reporting cadaveric OA diameters (approx 1.54mm) larger than modern angio means.

[36] C. Reis, et al., Anatomical variations of the ophthalmic artery: A study using computed tomography angiography, Experimental and Therapeutic Medicine 21, provides modern in-vivo CTA means (1.38mm) for comparison. (2021).

Figure Legends

Figure 1. Architecture of the entire pipeline. The pipeline is composed of three stages: the Prefiltering Stage, Segmentation Network, and Geometric Center Estimation.

Figure 2. Architecture of the Recurrent U-Net (R-UNet). The ConvGRU at the bottleneck maintains a spatial hidden state across sequential B-scans to ensure temporal consistency of the segmented boundaries.

Figure 3. (a) Microscope view of the OA section scanned by the OCT; the example slice is shown in red. (b) Geometric reconstruction process: A second-degree polynomial (blue) is fitted to the segmented boundary (red). The radius of curvature is used to define an osculating circle, the center of which (magenta) represents the estimated anatomical vessel center.

Figure 4. Box and whisker plots comparing the estimated vessel diameters against literature ground truth ranges. Note the tighter variance in the ICA and ACA compared to the OA.

Table Legend

Table 1. Comparison of Estimated Vessel Diameters vs. Literature Ground Truth.